

Problem Statement 2: Building a Traceable Waste Flow Network

Overview

Every day, waste flows through multiple invisible layers before it reaches recycling systems.

A typical journey looks like this:

- Waste is generated at the household
- Municipal systems collect and dump it at aggregation points
- Informal workers (ragpickers) recover valuable materials
- These materials move through middlemen (kabadiwalas)
- Finally, they reach recyclers or small-scale industries

Despite this active ecosystem, the system lacks structure, visibility, and efficiency.

Core Challenges

The current waste ecosystem is fundamentally broken because:

- There is no end-to-end tracking of waste movement
- Material flow is opaque and undocumented
- High value waste is lost due to poor sorting and inefficient routing
- Industries have no visibility into available recyclable supply
- Informal workers remain disconnected from structured demand systems

Today, waste moves through a fragmented chain—without a reliable chain of data.

The Challenge

Design a system that transforms this fragmented ecosystem into a **traceable, data-driven waste network**.

Your solution should:

- Enable digital tracking of waste movement across all stages
- Create visibility across stakeholders (collectors, aggregators, industries)
- Integrate informal workers without disrupting their existing workflow

- Improve efficiency in sorting, routing, and reuse of materials
 - Blockchain or distributed ledgers for tamper-proof tracking
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Technical Considerations

Focus on building system intelligence, not just a marketplace or UI.

Key questions to think about:

- How will you track waste units?
(*e.g., batch IDs, QR codes, weight-based logs*)
 - How will data be captured at each stage under low-tech constraints?
 - What kind of system architecture will you design?
(*centralized vs distributed systems*)
 - How will you ensure data authenticity and trust across stakeholders?
 - How can your system enable efficient matching between supply and demand?
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Evaluation Criteria

Your solution will be evaluated based on:

1. Problem Understanding

Depth of insight into real-world waste flows and challenges

2. System Design & Core Logic

Strength of architecture, tracking mechanisms, and decision-making logic

3. Innovation with Practicality

Feasibility of implementation within existing informal ecosystems